



# The impact of artificial intelligence adoption degree on corporate digital technology innovation

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## ABSTRACT

This study investigates the interplay among the AI adoption degree, dynamic capability, and industry concentration on corporate digital technology innovation. The research sample consists of manufacturing companies listed on the Shanghai and Shenzhen A-shares market. Machine learning methods are employed. The heterogeneity analysis reveals that sample companies can be categorised into two types. Additionally, decision tree analysis indicates that in intelligent enterprises, the impact of AI adoption degree on digital technology innovation is complex, while the absorbing ability positively influences digital technology innovation. In flexible enterprises, AI adoption degree, industry concentration, and absorbing ability all have a positive impact.

## ARTICLE HISTORY

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## 1. Introduction

The fusion of robotics with Internet of Things and big data has facilitated the gradual infiltration of artificial intelligence (AI) into the fabric of daily life. Omnipresent AI is profoundly reshaping both our personal routines and professional methodologies. Its applications span a broad spectrum, from the convenience of intelligent assistants to the innovation of autonomous vehicles, exerting a substantial influence across diverse industries (Sathikh, Fang, and Tan 2022). AI not only enhances production efficiency, but also offers a more personalised service experience. The new generation of information technology, epitomised by AI, is poised to serve as a crucial technical support and driving force for China's pursuit of high-quality economic development and Chinese-type modernisation. The primary objective of AI is to foster high-quality economic development, particularly as a vital catalyst for the digital economy. Notable AI technologies, such as industrial robots, have changed the way the economy operates (Haenlein and Kaplan 2020), and catalysed a fresh wave of social transformation.

Simultaneously, omnipresent AI also poses considerable challenges for businesses. A critical concern is the impact of AI on employment, with the China International Development Knowledge Center's 2022 Global Development Report suggesting that an estimated 85 million jobs worldwide may be displaced by automation between 2020 and

2025). Furthermore, AI technology fundamentally alters the decision-making and innovation processes within organisations (Cockburn, Henderson, and Stern 2019; Wang, Xiao, and Deng 2023). In the manufacturing sector, AI enhances decision-making efficacy and employee efficiency, consequently boosting overall organisational productivity and reducing operational costs (Chen et al. 2020; Duan, Edwards, and Dwivedi 2019). In marketing, AI facilitates more effective analysis of customer data, thereby improving customer satisfaction and market competitiveness (Caldarini, Jaf, and McGarry 2022). Additionally, the advent of AI has revolutionised the innovation process and the dynamics of research and development within companies (Cockburn, Henderson, and Stern 2019). AI has accelerated the digital innovation process by blurring the traditional boundaries between various stages of innovation – discovery, development, diffusion, and impact – resulting in less clarity regarding the initiation, conclusion, and progression in time and space (Nambisan et al. 2017). Consequently, the interdependence between the innovation process and its outcomes has become increasingly complex and dynamic (Nambisan et al. 2017). In the AI era, the decision to adopt AI, as well as the extent of the implementation, critically influences an enterprise's operational sustainability and overall success or failure.

In the current era of globalisation and rapid advancements in science and technology, the digital economy has emerged as a pivotal force driving economic growth, capturing the attention of both governments and enterprises. Digital technological innovation serves as the primary catalyst for the development of the digital economy. It not only provides traditional enterprises with a pathway for digital transformation and high-quality development, but also plays a crucial role in enhancing the competitiveness and influence of the digital economy. Digital technology innovation has become the most integrated and widely applicable element within the technological innovation landscape, representing the strongest driving force in this field. As the micro-foundation of economic operations, enterprises are the key agents in fostering digital technology innovation. As a critical strategic resource, the adoption of AI technology directly influences resource allocation and utilisation within enterprises, thereby significantly enhancing their capabilities for digital technology innovation. AI technology dismantles the barriers of traditional industries, encouraging enterprises to pursue cross-industry collaborations for resource sharing and synergistic partnerships, which collectively propel advancements in digital technology innovation. Therefore, exploring the impact of AI adoption on enterprises' digital technology innovation is of significant importance. Such study will facilitate enterprises in bolstering their digital technological creativity, thereby contributing to the enhancement of the nation's overall digital innovation ability.

Existing literature has examined the relationship between AI adoption and innovation. Trocin et al. (2021) proposed a framework for AI-enabled digital innovation, elucidating how the implementation of AI functions facilitates digital process and service innovations within organisations. Sjödin et al. (2021) conducted an exploratory multi-case study involving six industrial companies focused on digital services, revealing that businesses can achieve model innovation by developing three specific AI capabilities. Research by Wang, Xiao, and Deng (2023) on China's manufacturing sector indicates that the use of industrial robots significantly fosters innovation. The marginal effect of industrial robot implementation is more pronounced in capital-intensive and high-tech industries than in labour-intensive sectors. Furthermore, its impact on substantive innovation is greater

than on strategic innovation. Additional studies support the influence of AI adoption on product innovation (Mariani and Fosso Wamba 2020) and process innovation (Chatterjee et al. 2021). In terms of research objects, although existing research has enhanced the discourse on AI adoption and innovation, the connection between AI adoption and digital technology innovation remains underexplored. In terms of research paradigm, the prevailing research predominantly relies on qualitative case studies or theoretical analyses, with only a limited number of studies employing large-sample empirical methods to investigate the factors influencing digital technology innovation. Furthermore, most existing research emphasises the net effects between variables, resulting in insufficient exploration of the complex nonlinear relationships involved.

While AI confers significant benefits to businesses, it also introduces a variety of practical challenges. Traditional manufacturing enterprises, for example, usually grapple with challenges such as the lack of standardised authoritative data and the lack of unified management system architecture, which can lead to a digital paradox. This raises the question: Does a high AI adoption degree necessarily correlate with enhanced digital innovation? Many enterprises may lack the necessary conditions and capabilities to effectively leverage digital technologies. Consequently, even with proactive adoption of artificial intelligence, they may struggle to generate significant value. Thus, it is crucial to examine the interaction between AI adoption degree and additional influencing factors, as well as the synergistic effects these interactions have on digital technology innovation. This paper proposes a research model based on the Technology-Organization-Environment (TOE) framework and employs machine learning methods to investigate the following research questions: (1) What are the key drivers of enterprise digital technology innovation within the TOE framework? (2) Can enterprises be classified into distinct categories based on characteristic variables, and what characteristics do these various types exhibit? (3) For these various enterprise types, how does the AI adoption degree affect digital technology innovation? Additionally, what complex nonlinear relationships exist concerning the synergistic effects of characteristic variables on digital technology innovation?

This paper contributes to the research in several ways: Firstly, it employs the TOE framework to develop a research model, thereby broadening the applicability of the framework. Secondly, the study introduces a novel categorisation of enterprises based on their characteristics, examining the complex and nonlinear effects of AI adoption degree and the interplay among these characteristics on digital technology innovation across different enterprise types. This nuanced perspective enriches our understanding of the precursors to digital technology innovation and enhances the body of research related to innovation theory. It also provides a more granular view of how AI adoption can be tailored to the specific needs and capabilities of different types of enterprises. Thirdly, the findings offer practical insights for businesses on the strategic use of AI. By revealing the conditions under which AI can be effectively leveraged to enhance digital innovation, our research aids organisations in making informed decisions about the allocation of resources and capabilities. This is particularly pertinent in the current digital landscape, where the competitive edge is commonly determined by the ability to innovate rapidly and efficiently.

The arrangement of this study is structured as follows: [Section 2](#) provides a thorough review of existing literature about AI adoption and its influence on digital technology

innovation. [Section 3](#) outlines the research design, detailing feature index selection, the measurement of enterprise digital technology innovation, research methods and research ideas, and data sourcing, acquisition, and processing. [Section 4](#) presents the analytical procedures and findings, which include an analysis of variable interrelations, the calibration of variables, the application of cluster analysis, the deployment of decision analysis techniques, and the evaluation of decision rules. Finally, [Section 5](#) provides the research conclusion and enlightenment, including research conclusion, practical implications, and limitations and future research directions.

## 2. Literature review

### *2.1. AI adoption degree and enterprise digital technology innovation*

The notion AI, initially conceptualised as the science and engineering focused on creating intelligent technologies, particularly intelligent computer programs. As of now, there remains no unified definition of AI in academia despite its growing prevalence, academia still grapples with a singular, comprehensive definition of AI (Duan, Edwards, and Dwivedi 2019). Current definitions of AI are predominantly categorised into two perspectives. On the one hand, starting from the underlying technology, AI is defined as the technology that enables computers to simulate human intelligence using logic, hypothetical rules, and machine learning (Prem 2019). On the other hand, regarding functional outcomes, artificial intelligence is described as the capacity of computers to perform tasks – such as visual perception, speech recognition, and decision-making – that typically necessitate human intelligence (Larson and Dechurch 2020). The adoption of technology represents an ongoing dynamic process between organisations and individuals. In this context, organisations make informed, context-specific decisions regarding technology investments, implement the technology, and individuals subsequently utilise and adapt these technologies. From an organisational perspective, the adoption of artificial intelligence technology means that organisations are aware of the advantages that artificial intelligence technology brings and are beginning to acquire the resources associated with it (Alsheiabni, Cheung, and Messom 2019). The AI adoption degree reflects the depth and breadth of this process. Although various organisations are in the process of adopting AI, empirical research on this topic remains limited.

The existing literature has extensively examined the influence of the digital economy on the operations of micro-enterprises, particularly the field of enterprises digital transformation (Aghakhani et al. 2024; Tian and Ou 2024). Digital technology is the technical support for enterprises digital transformation, with its innovation process being pivotal to the effectiveness of digital transformation (Shi, Mai, and Wu 2022). The proliferation and extensive adoption of digital technology have fundamentally transformed the nature of innovation. This transformation influences innovation behaviours, the integration of innovative resources, and the advancement of innovation processes and business models (Nambisan, Wright, and Feldman 2019). The existing literature presents various definitions of digital technology innovation. However, most literature has a consistent understanding of digital technology innovation: digital technology innovation refers to the process and results of enterprises or organisations creating new products, processes, organisations, and business models based on digital technology (Nambisan et al. 2017). Within this

context, the term ‘innovation’ emphasises both the process and the outcomes of innovation, with patent output often serving as a tangible indication of such innovation in numerous studies. Digital technologies encompass a range of components, including information, computing, communication, and connectivity technologies, as well as their combinations – such as AI, big data, cloud computing, and blockchain (Vial 2019). These emerging digital technologies are increasingly integrated into the production and operational activities of various traditional enterprises. They have the potential to replace conventional human labour, optimise production processes, and reconfigure traditional production factors, thereby altering established value creation models (Acemoglu and Restrepo 2018). In recent years, scholars have made significant advancements in the field of digital innovation, exploring new mechanisms, management frameworks, and innovation ecosystems.

AI is a pivotal digital technology that empowers enterprises to advance and thrive in the digital era, significantly contributing to innovation (Kakatkar, Bilgram, and Füller 2020). Current research predominantly examines the role of AI adoption in fostering product innovation within enterprises (Mariani and Fosso Wamba 2020). Additional studies have explored the implications of AI adoption on process innovation (Chatterjee et al. 2021) and business model innovation (Burstrom et al. 2021). However, there is a scarcity of research addressing the AI adoption degree in relation to digital technology innovation. The AI adoption facilitates innovation analytics (Mariani and Nambisan 2021), allowing enterprises to harness and enhance product innovation. Furthermore, AI adoption and analytics contribute to mitigating risks associated with innovation while increasing innovation objectives (Mariani, Machado, and Nambisan 2023). Its implementation bolsters enterprises’ capabilities to restructure and redesign process innovation (Chatterjee et al. 2021). Numerous successful organisations are currently integrating AI into their operations, resulting in the development of distinctive AI-driven business models (Anton et al. 2021). Burstrom et al. (2021) employed a qualitative methodology to investigate four large, established global manufacturing firms undergoing transformation through AI adoption. Their findings indicate that AI adoption significantly influences the processes of value creation, value delivery, and value capture, thereby facilitating business model innovation.

AI adoption introduces innovative methodologies for information processing, leveraging its computational power to enhance data analysis, thereby accelerating the discovery process and facilitating the development of new solutions. This trend is driving a fresh wave of digital innovation. AI systems can learn from extensive datasets, recognise patterns, and generate predictive insights. A multi-case study conducted by Trocin et al. (2021) demonstrates that the AI adoption positively influences digital innovation processes, digital service innovation, and the outcomes of digital innovations. Additionally, an exploratory multi-case study by Sjödin et al. (2021) involving industrial enterprises and their digital partners reveals that manufacturers must cultivate three interrelated AI capabilities – data pipeline capabilities, algorithm development capabilities, and AI democratisation capabilities – to adequately capitalise on digital servitization.

In terms of research subjects, current research has enriched the discussion on the relationship between the AI adoption degree and innovation. However, there remains a lack of comprehensive exploration regarding the relationship between AI adoption and digital technology innovation. Furthermore, the majority of existing studies

predominantly examine the net effect of AI adoption on enterprise innovation, often overlook the intricately linked effects of various factors alongside AI adoption on digital technology innovation. In the absence of sufficient conditions and capabilities to effectively implement digital technologies, enterprises may struggle to generate substantial value, even with active adoption of AI technologies. Consequently, it is essential to examine the interactive effects of other factors alongside the AI adoption degree and to investigate the synergistic impacts on digital technology innovation. In terms of research paradigm, the majority of existing studies on this topic are predominantly case studies or theoretical analyses. Only a limited number of studies employ large-sample empirical analysis to investigate the factors influencing digital technology innovation. Furthermore, most of the current literature primarily examines the net effects of variables, while offering insufficient exploration of complex nonlinear relationships.

## **2.2. Model construction**

Based on the preceding analysis, it is crucial to examine the influence of AI adoption on enterprise digital technology innovation. However, evaluating solely the technical factors is an overly narrow perspective, a comprehensive analysis must incorporate additional dimensions. The TOE framework categorises the factors influencing technological innovation implementation in enterprises into three levels: technological, organisational, and environmental (Tornatzky and Fleischer 1999). This framework provides a robust theoretical foundation for investigating enterprise technology innovation and adoption (Kimiagari and Baei 2021) and is extensively utilised in information systems research. The specific influences of technology, organisation, and environment vary across contexts, thus the determination of the influence factors should be determined according to the specific research situation. In this paper, the characteristic variables influencing digital technology innovation are extracted in accordance with the TOE framework.

The AI adoption degree pertains to the technical level and exerts a positive influence on both the digital innovation process and its outcomes (Trocin et al. 2021). At the organisational level, the theory of dynamic capability explores how enterprises can integrate, develop, and reconfigure internal and external resources and capabilities to foster new competencies necessary for adapting to a rapidly evolving environment. Dynamic capability confers competitive advantages (Teece 2014), and helps enterprises to find opportunities and implement digital technology innovation. The term 'dynamic' in dynamic capability underscores the critical role of organisations in persistently enhancing or redefining their competitive edge to align with a constantly evolving external environment. Conversely, the term 'capabilities' emphasises the significance of organisations in effectively managing, integrating, and restructuring both internal and external knowledge, resources, and skills to adapt to these environmental changes. A enterprise with strong dynamic capability is able to recognise opportunities for digital innovation as well as the latest advancements in digital technologies, including AI, big data, and cloud computing. Furthermore, this enterprise is also adept at evaluate the potential impact of these technologies on its business model. Upon identifying an opportunity, the enterprise leverages its dynamic capability to promptly realign its internal resources to facilitate the implementation of digital technological innovations. Concurrently, in a complex business environment, market demand and competition present significant

challenges for innovation management. As a significant characteristic of market structure, industry concentration influences a firm's innovation capability to a certain extent. In industries characterised by low concentration, digital technology plays a critical role in enhancing innovation. Conversely, in highly concentrated industries, digital technology emphasises the quality of innovations. In highly concentrated industries, a limited number of companies possess significant market resources and capital. These resources facilitate substantial investments in digital technology innovation, including research and development funding, recruitment of highly skilled personnel, and acquisition of advanced technologies. The concentration of resources enables these companies to undertake larger-scale and more extensive digital technology innovation initiatives.

Simultaneously, a synergistic interaction exists among the various features. AI adoption degree is mutually reinforced by dynamic capability. Enterprises with robust absorbing ability can efficiently integrate and utilise external knowledge and information, making them tend to embrace new technologies such as AI. Furthermore, adaptive ability and innovation ability empower enterprises to effectively leverage new technologies, thereby enhancing the output of digital technology innovations following the implementation of AI. Meanwhile, the AI adoption can improve organisational efficiency (Yang 2022), optimise production processes, and identify potential market opportunities through data analysis. This enhances enterprises' flexibility, enabling them to respond more effectively to market changes and diverse customer needs, thereby fostering the development of dynamic capabilities. Industry concentration also influences both AI adoption degree and dynamic capabilities. In industries characterised by low concentration and heightened competition, resources tend to be dispersed, and enterprises may experience a deficiency in funding and technical support. To survive and thrive in fierce market competition, enterprises may adopt a more cautious approach, and resort to conservative strategies. As a result, these enterprises may exhibit reluctance in embracing new technologies, particularly those with high investment and high risk, such as AI. This hesitance ultimately leads to a lower rate of AI adoption degree. In industries characterised by low concentration, enterprises may experience reduced motivation to innovate due to intense competition. Additionally, they may face challenges in identifying and capitalising on market opportunities due to resource constraints, which can hinder the enhancement of their dynamic capabilities.

In summary, this paper investigates the influence of the AI adoption degree on enterprise digital technology innovation at a technical level. Furthermore, it examines the effects of the synergistic interaction among dynamic capability at the organisational level, industry concentration at the environmental level, and AI adoption degree on enterprise digital technology innovation. The research framework is illustrated in [Figure 1](#).

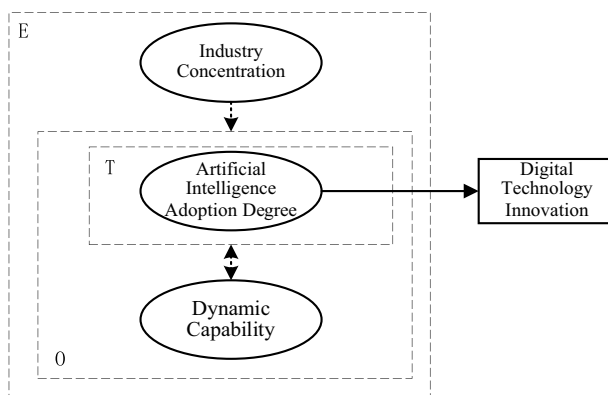
### 3. Materials and methods

#### 3.1. Feature index selection

##### 3.1.1. AI adoption degree (AIAD)

Some scholars evaluate an enterprise's AI orientation by analysing the proportion of AI-related keywords found in the annual reports of publicly listed companies. Others assess the intensity of AI technology within an enterprise by examining the quantity of effective





**Figure 1.** Research framework based on TOE.

AI patents. The frequency of keywords associated with AI indicates the level of attention an enterprise dedicates to AI, whereas AI patents represent the enterprise's R&D capability. Within the context of manufacturing production, there is a complex interplay between workforce size, employment composition, and labour earnings, all of which are closely associated with the degree of AI adoption. Building on the work of He et al. (2020), this study adopts a per capita measure of machinery and equipment value as a proxy for assessing the extent of AI adoption degree within enterprises. This metric is calculated by dividing the book value of machinery, as detailed in the fixed assets section of the company's annual financial statements, by the total workforce count, thereby providing an insightful indicator of AI adoption within the manufacturing sector. Specifically, we obtain the annual reports of publicly listed companies from CNINFO to acquire the book value of machinery for these enterprises. Additionally, the number of employees for the publicly listed companies are sourced from the China Stock Market & Accounting Research Database (CSMAR).

### 3.1.2. *Dynamic capability (DC)*

Dynamic capability refers to an organisation's ability to comprehensively reconfigure or integrate its internal and external resources in response to environmental changes. This concept is inherently complex and multidimensional. Notably, existing literature on measuring dynamic capability predominantly relies on survey questionnaire design scales. This approach may pose challenges related to data collection, including issues such as inadequate questionnaire design and potential misinterpretation by respondents. In this paper, we adopt the methodology of Zhao, Wang, and Wang (2016) to assess dynamic capability through three dimensions: innovation ability, absorbing ability, and adaptive ability. The data necessary for assessing dynamic capability is obtained from the CSMAR. The specific procedures are outlined below.

**3.1.2.1. Innovation Ability (IA).** Innovation ability refers to an enterprise's ability to continuously integrate resources, enhance its organisational structure, and update its products, technologies, and services. Refer to existing literature (Zhao, Wang, and Wang



2016), this study employs two indices – annual R&D expenditure intensity and the proportion of technical personnel – to comprehensively assess the innovation ability of the sample company. The data for each of these indexes is standardised individually, and the overall innovation ability is derived by summing these standardised values, as represented in formula (1):

$$IA = \frac{(X_{RD} - \min_{RD})}{(\max_{RD} - \min_{RD})} + \frac{(X_{rr} - \min_{rr})}{(\max_{rr} - \min_{rr})} \quad (1)$$

Where  $RD$  stands for R&D expenditure intensity and  $rr$  stands for the proportion of technical personnel.

**3.1.2.2. Absorbing Ability (ABA).** Absorbing ability pertains to a company's ability to recognise and acquire valuable knowledge and information and subsequently internalise it within the organisation through processes of comprehension and assimilation. Refer to existing literature (Zhao, Wang, and Wang 2016), the R&D expenditure intensity, defined as the ratio of annual R&D expenditures to operating income for the sampled firms, was utilised to assess absorbing ability.

**3.1.2.3. Adaptive Ability (ADA).** Adaptive ability refers to an organisation's ability to reallocate resources in order to respond flexibly when the external environment changes. Refer to existing literature (Zhao, Wang, and Wang 2016), the adaptive ability of a firm is assessed through the coefficient of variation of its annual R&D expenditure, capital expenditure, and advertising expenditure. This coefficient is defined as the ratio of the standard deviation to the mean of these three indicators. A smaller coefficient of variation indicates a stronger adaptive ability, implying that the firm can better withstand changes in the external environment. To align the value of the coefficient of variation with adaptive ability, negative values are applied to the coefficient. Consequently, a larger adjusted coefficient of variation signifies a stronger adaptive ability of the enterprise.

### 3.1.3. Industry concentration (IC)

Industry concentration is commonly used to reflect the industry competition degree, and its main indicators are the Herfindahl-Hirschman Index ( $HHI$ ) and the Concentration Ratio 4 ( $CR4$ ). The  $HHI$  not only indicates the relative size of firms within an industry but also captures the distribution of firm sizes. In the study conducted by Lin, Zhu, and Chen (2021), the  $HHI$  is employed to assess industry concentration, as illustrated in Formula (2):

$$HHI = \sum \left( \frac{X_i}{X} \right)^2 \quad (2)$$

Where  $X_i$  is the revenue or assets of enterprise  $i$ , and  $X$  is the revenue or assets of the entire industry.  $HHI$  is a comprehensive index that measures industry concentration. A higher index indicates reduced dispersion in the distribution of industry revenue or total assets. Consequently, increased industry concentration is associated with diminished market competition intensity.

The data required to assess industry concentration is sourced from the CSMAR. In this paper, the industry concentration is defined as the sum of the squared proportions of

each enterprise's primary business income relative to the total primary business income within different industries. This relationship is represented by Formula (3):

$$IC = \sum \left( \frac{\text{The main business income of enterprise } i}{\text{The main business income of the whole industry}} \right)^2 \quad (3)$$

### 3.2. Measurement of corporate digital technology innovation

The number of patent applications serves as a common index for assessing the level of R&D innovation within enterprises. Prior studies have utilised patent text information to identify digital patents at the enterprise level, measuring digital technology innovation based on the volume of patent applications (Liu et al. 2023). The patent data utilised in this paper are sourced from the patent database of the State Intellectual Property Office of China (SIPO). Match the patent's IPC classification number with the digital technology patent classification number in the Key Digital Technology Patent Classification System (2023). This alignment allows for the determination of whether a given patent qualifies as a digital patent. Consequently, the study quantifies the number of digital invention patent applications and digital utility model patent applications filed by enterprises. The overall count of digital patent applications for each enterprise is derived by summing the digital invention and utility model patent applications. The number of digital patent applications of enterprises is added by 1 and the natural logarithm is taken to measure the digital technology innovation of enterprises, which is denoted as DTI.

### 3.3. Research methods

Previous literature tends to treat research subjects in a generalised manner, which can limit the universal applicability of its findings. Conducting a cluster analysis of research subjects and focusing on the relationships between variables in varying contexts can mitigate these challenges. While many studies rely on multiple linear regression models to isolate the 'net effect' of individual variables, this approach can overlook the complex, real-world interactions between multiple factors that contribute to outcomes. Typically, the relationships among variables exhibit nonlinear and complex characteristics (Zhou and Li 2023). For instance, traditional manufacturing enterprises may face significant challenges, such as the lack of standardised data and a cohesive management system architecture, leading to a 'digital paradox'. Even with proactive AI adoption, companies lacking the necessary conditions and capabilities might struggle to derive significant value from digital technologies. Thus, it is essential to consider the interplay between various factors and the AI adoption degree, as well as their synergistic effects on digital technology innovation. Decision tree analysis is particularly well-suited for investigating complex nonlinear relationships between variables. By applying decision tree analysis to formulate decision rules, we can gain intuitive insights into the conditions that correspond with high and low levels of digital technology innovation. This paper employs a comprehensive approach, utilising cloud model calibration, cluster analysis, decision tree analysis, and other machine learning methods to explore the following research questions. What are the key driving factors behind corporate digital technology innovation within the TOE framework? Can enterprises be classified into distinct categories

based on characteristic variables, and what characteristics do these various types exhibit? Moreover, for various enterprise classifications, how does the AI adoption degree influence digital technology innovation? Additionally, Is there a complex nonlinear relationship between characteristic variables and digital technology innovation? The research framework is illustrated in Figure 2.

Initially, the driving factors of digital technology innovation are organised according to the TOE framework, from which the research indicators for this study are identified. Subsequently, the cloud model is employed to calibrate the original variable data. The research data is sourced from a database. To mitigate the effects of dimension and extreme values, the cloud model is utilised to standardise the original variable data, resulting in variable values normalised within the range of 0 to 1. D. Li (2000) introduced the cloud model, a method that facilitates the conversion of qualitative concepts into quantitative expressions, which is reflective of the inherent fuzzy and random present in human cognitive processes. This approach offers innovative perspectives for the examination of uncertainties in AI. The cloud model integrates fuzzy membership degree with random data, transforming qualitative concepts into quantitative values through a cloud generator, thus achieving a ‘soft division’ of evaluation indicators.

Additionally, a clustering algorithm was employed to categorise the calibrated data into distinct groups. The clustering process aims to maximise the aggregation of similar data objects while separating different kinds of data objects as much as possible, thereby enhancing the focus of subsequent research efforts. Moreover, clustering serves as a control variable by grouping samples with shared characteristics into a single category, effectively managing variance among the samples. The K-Means clustering algorithm partitions the dataset into K distinct clusters, each formed around data points that exhibit minimal internal distance (Celebi, Kingravi, and Vela 2013). The optimal number of clusters (K) can be determined using the elbow method.

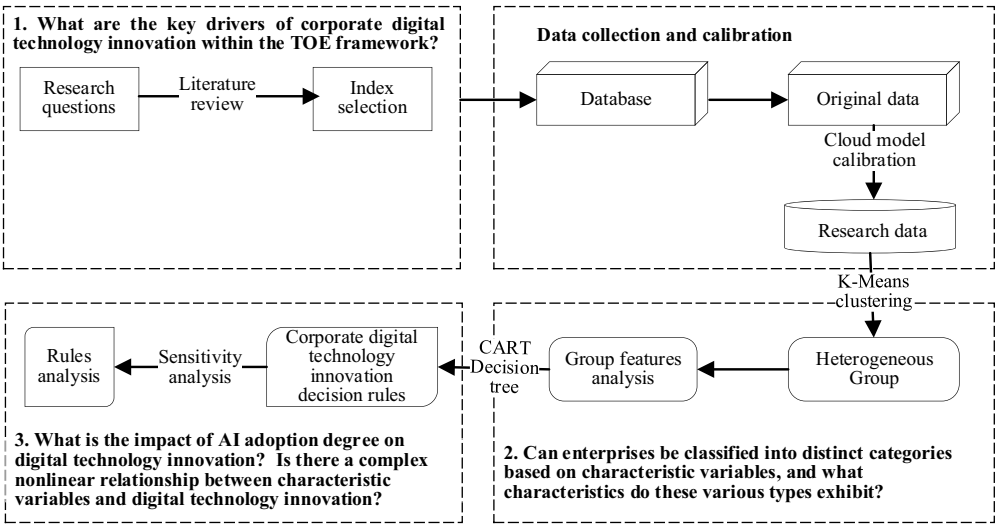


Figure 2. Research framework.

Finally, the decision tree analysis method is employed to investigate the intricate nonlinear relationships between characteristic variables and digital technology innovation across different groups. Specifically, the decision tree is constructed by establishing the relevant parameters and pruning strategies. Sensitivity analysis is conducted based on the *Support* and *Confidence* of the decision tree branches (Zhou and Li 2023). By extracting the decision rules of interest from the decision tree, the interaction effect of the nonlinear complex influencing factors of the target dependent variable is analysed. The Classification and Regression Trees (CART) algorithm generates a relatively concise tree structure characterised by its binary tree structure, resulting in rules that are straightforward and easily interpretable. The core nodes of the decision tree signify the evaluations of specific attributes, while the leaf nodes indicate the final classifications. Pruning methods are utilised to mitigate the complexity of the tree structure. CART decision tree pruning strategies are categorised into pre-pruning and post-pruning methods. Pre-pruning involves establishing thresholds for tree depth and the minimum number of samples at the leaf nodes. In contrast, post-pruning occurs after the construction of the decision tree, wherein nonleaf nodes are examined from the bottom upward. If the performance can be enhanced by converting the subtree associated with a node into a leaf node, the subtree is replaced accordingly. The *Support* and *Confidence* of the branches in the decision tree are computed as follows.

The *Support* is defined as the ratio of the number of samples within a specific rule to the total number of samples, as indicated in formula (4):

$$Support_i = \frac{D_i}{D} \times 100\% \quad (4)$$

Where  $Support_i$  is the *Support* for rule  $i$ ,  $D_i$  is the number of samples for rule  $i$ , and  $D$  is the total sample size.

The *Confidence* is the conditional probability of a rule occurring, as shown in formula (5):

$$Confidence_i = \frac{C_i}{C} \times 100\% \quad (5)$$

Where  $Confidence_i$  refers to the *Confidence* of rule  $i$ ,  $C_i$  refers to the number of samples that support the rule  $i$ , and  $C$  refers to the total sample size of rule  $i$ .

### 3.4. Data source, acquisition and processing

This study selects manufacturing listed companies from Shanghai and Shenzhen A-shares as the initial sample. The characteristic variable data pertains to the year 2021, while the data on digital technology innovation corresponds to the year 2022. The data sources for the sample companies include: (1) Basic information and financial indicators obtained from the China Stock Market & Accounting Research Database (CSMAR). (2) Annual report texts come from CNINFO. (3) Patent data sourced from the Patent Database of the State Intellectual Property Office of China (SIPO). The data processing involved the following steps: (1) Elimination of samples

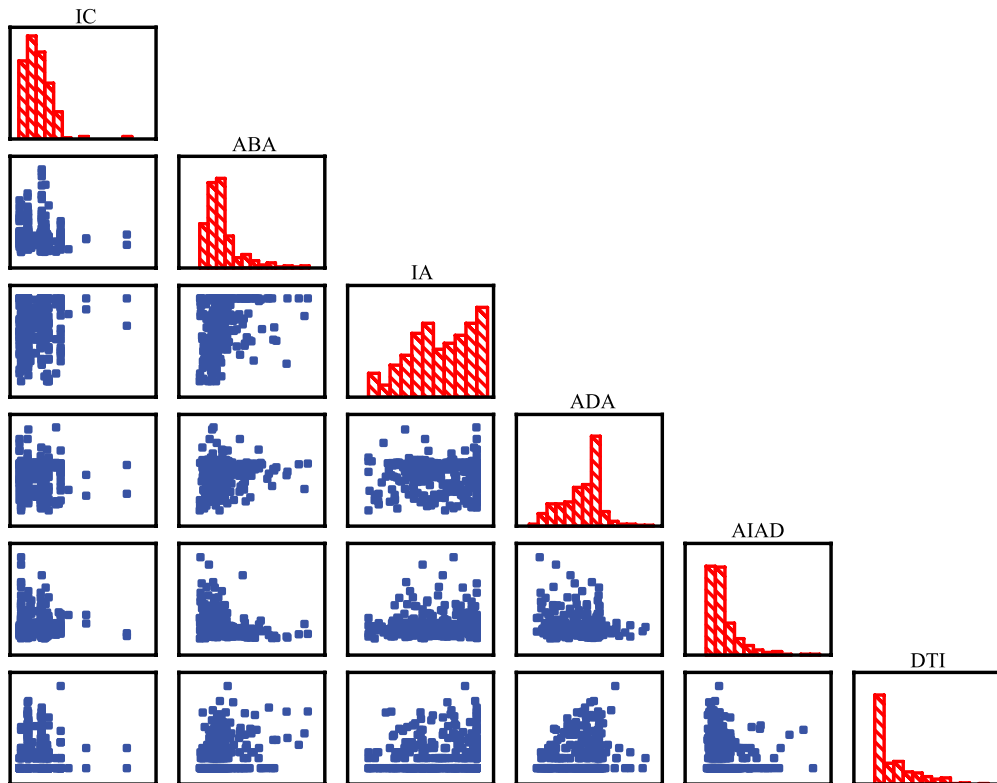
with missing key variables. (2) Exclusion of samples exhibiting business anomalies (stock codes ST, \*ST, or PT). Ultimately, data from 288 companies was retained for analysis.

## 4. Results

### 4.1. Correlation analysis

The purpose of correlation analysis is to mitigate the impact of highly correlated variables on research outcomes. The Pearson correlation coefficients were computed among six variables: AI adoption degree, industry concentration, absorbing ability, innovation ability, adaptive ability, and digital technology innovation. Additionally, scatter plots and histograms depicting the frequency distribution of characteristics for each pair of variables were created, as illustrated in Figure 3.

As illustrated in Figure 3, there is no evident linear relationship among the variables, suggesting the absence of multicollinearity among the feature indicators. A weak positive correlation is observed between absorbing ability and digital technology innovation, indicating that digital technology innovation is influenced by multiple factors rather than a single determinant. The nonlinear relationships within the data can be effectively analysed using machine learning methods.



**Figure 3.** Scatter plot and histogram of characteristic frequency distribution of correlation analysis before calibration.

## 4.2. Variable calibration

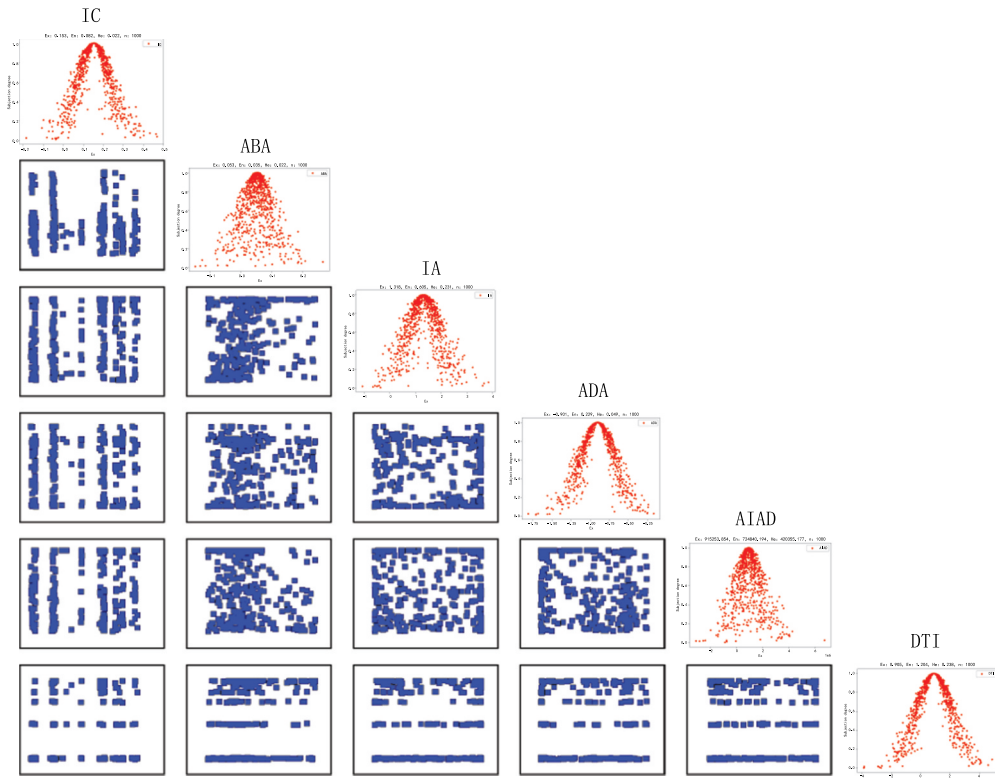
In the objective world, the development of processes and the distribution of data typically exhibit normal distribution characteristics. However, as demonstrated in the histogram of frequency distribution presented in Figure 3, except for adaptive ability, the distribution characteristics of other variables are biased and do not obey the strict normal distribution. Additionally, each characteristic variable possesses varying values, and direct analysis using the original data may lead to dominance by variables with larger values, obscuring the impact of others. To mitigate these issues, it is essential to calibrate the data to improve comparability among the variables. Current calibration methods overlook the inherent uncertainties associated with the original system features and fail to effectively address noise. Furthermore, most calibration processes merely involve linear or nonlinear transformations of the current state, essentially functioning as compression and correction of the data range or distribution. Consequently, these methods are significantly constrained by the uncertainty characteristics of the system data. Cloud model is a mathematical framework designed address uncertainty. It characterises the uncertainty associated with a concept or data through three numerical features: Expectation ( $Ex$ ), Entropy ( $En$ ), and Hyper-Entropy ( $He$ ). This model provides a practical approach for the conversion of qualitative concepts to quantitative results (D. Li 2000). The cloud model calibration involves three core processes: cloud digital feature extraction, cloud drop generation, and membership function transformation (H. Li et al. 2024). We employ the cloud model method for data calibration, transforming the data to a normalised scale between 0 and 1. The thresholds for constructing the cloud model are established at the 75% quantile for the upper threshold and the 25% quantile for the lower threshold, as outlined in Table 1. We posit that data falling within the interquartile range aligns more closely with the fundamental characteristics of normal distribution, whereas values outside this range are often excessively high or low, resulting in diminished explanatory power for management decision-making compared to sample data situated near the mean.

To illustrate the distribution of the calibrated data, a scatter plot depicting the relationships among the calibrated variables, along with a feature cloud model diagram, was constructed, as presented in Figure 4.

Figure 4 illustrates that post-calibration, the distribution of variables becomes more uniform. This indicates that employing a cloud model for data calibration effectively mitigates the influence of data bias on the analytical outcomes.

**Table 1.** Calibrate threshold points.

| Variable                            | High Threshold | Low Threshold |
|-------------------------------------|----------------|---------------|
| Industry Concentration (IC)         | 0.199          | 0.101         |
| Absorbing Ability (ABA)             | 0.060          | 0.030         |
| Innovation Ability (IA)             | 1.857          | 0.918         |
| Adaptive Ability (ADA)              | −0.729         | −1.063        |
| AI Adoption Degree (AIAD)           | 1 116,303.000  | 387 864.354   |
| Digital Technology Innovation (DTI) | 1.442          | 0.000         |



**Figure 4.** Calibrated correlation analysis scatter plot and cloud model plot.

### 4.3. Cluster analysis

To ascertain the optimal number of clusters, the average distortion degree for various cluster quantities was calculated using the elbow method. The results obtained from the elbow algorithm are presented in [Figure 5](#).

As illustrated in [Figure 5](#), when the number of clusters exceeds two, the variation amplitude of the average distortion decreases. Consequently, the optimal number of K-Means clusters is determined to be 2. The K-Means algorithm is employed to categorise enterprises into these two distinct groups. Subsequently, a clustering feature radar chart is generated based on the mean values of the characteristic indicators for each category of enterprises, as depicted in [Figure 6](#).

According to [Figure 6](#), two distinct types of enterprises can be identified. The first type, termed 'intelligent enterprises', exhibits a higher average AI adoption degree, coupled with a lower average across the three dimensions of dynamic capability. In contrast, the second type, referred to as 'flexible enterprises', demonstrates a lower average AI adoption degree while maintaining a relatively elevated average in the same three dimensions of dynamic capability. Notably, the average level of industry concentration remains statistically similar between the two enterprise types. [Table 2](#) presents the classification details, indicating that flexible enterprises are more numerous, with minimal variation observed in the proportion of different categories of digital technology innovation between the two types.



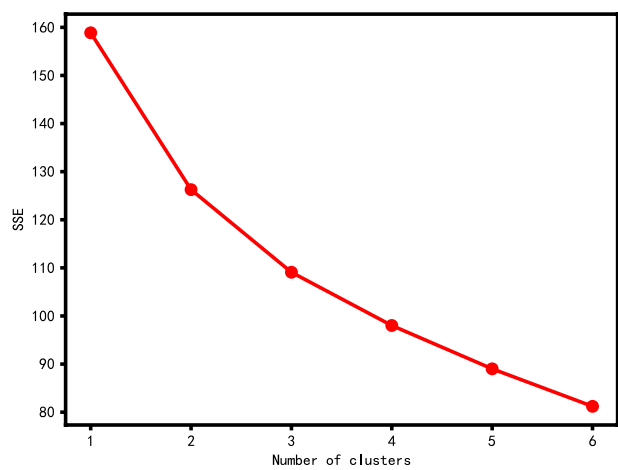


Figure 5. Elbow algorithm clustering number graph.

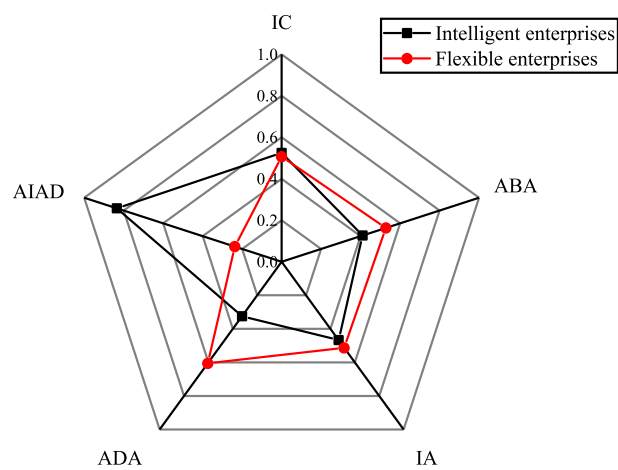


Figure 6. Radar map of clustering features.

Table 2. The features of the two clusters.

| Firm type               | Number of enterprises | IC    | ABA   | IA    | ADA   | AIAD  | Proportion       |
|-------------------------|-----------------------|-------|-------|-------|-------|-------|------------------|
| Intelligent enterprises | 136                   | 0.525 | 0.410 | 0.467 | 0.324 | 0.835 | l: 52%<br>h: 48% |
| Flexible enterprises    | 152                   | 0.507 | 0.528 | 0.513 | 0.605 | 0.237 | l: 49%<br>h: 51% |

4.4. Decision tree analysis

Decision trees are invaluable tools in both the theory and practice of management, with the CART algorithm being particularly esteemed for its ability to effectively identify interactions among variables. This study aims to delve deeper into the influence of feature indicators on digital technology innovation through cluster analysis. Five feature indicators are identified as conditional attributes, with the discretised levels of digital

technology innovation serving as decision attributes. The study imposes a maximum tree depth of three, mandates a minimum of ten samples per leaf node, and employs manual pruning to refine the model. This process ensures a clear understanding of how various feature combinations influence digital technology innovation. Within the decision tree, nonleaf nodes represent data distributions, while the leaf nodes illustrate the proportions of decision attribute categories, with the category exhibiting the highest proportion reflecting the resultant decision rules.

#### **4.4.1. Decision tree for intelligent enterprises**

In intelligent enterprises, the extent of AI adoption degree and the absorbing ability have an impact on digital technology innovation. Furthermore, varying combinations of features exert distinct effects on this innovation process. The decision tree for intelligent enterprises is illustrated in [Figure 7](#).

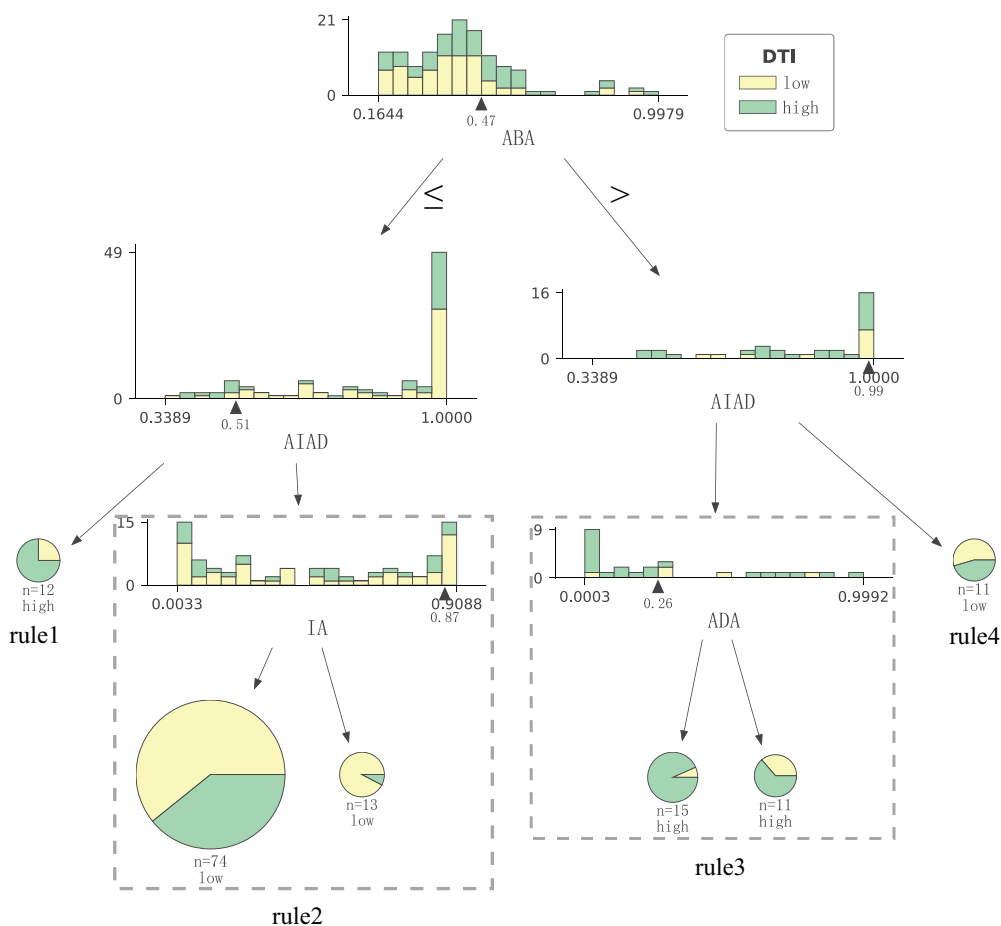
As illustrated in [Figure 7](#), the left subtree of the intelligent enterprises decision tree comprises two sub-nodes: absorbing ability and AI adoption degree. When the absorbing ability of intelligent enterprises is low, a reduction in the level of AI adoption degree may facilitate digital technology innovation. Excessive implementation of AI technologies can lead to ‘indigestion’ where even robust innovation ability fails to enhance digital technology innovation. The inability to further differentiate between high and low levels of digital technology innovation, despite innovation capability being a conditional attribute, suggests that the available information is insufficient for this distinction.

The right subtree consists of two nodes: absorbing ability and AI adoption degree. In intelligent enterprises, a high level of absorbing ability, coupled with an elevated degree of AI adoption degree, facilitates significant digital technology innovation. Conversely, excessive reliance on AI may lead to a lower level of digital technology innovation. The ineffectiveness of the adaptive ability condition in differentiating between high and low levels of digital technology innovation indicates its impact is weakened. This may be attributed to the fact that the AI adoption degree at the parent node already provides sufficient information to distinguish high levels of digital technology innovation. Comparative analysis reveals that, within intelligent enterprises, both high and low levels of digital technology innovation are influenced by absorbing ability and AI adoption degree, while the effects of innovation ability and adaptive ability remain not significant.

In general, a strong absorbing ability contributes to a high level of digital technology innovation in intelligent enterprises. However, an excessive pursuit of perfection and a high degree of AI adoption can lead to a decline in digital technology innovation. This can be attributed to the diminishing marginal returns associated with AI as a form of technological innovation input. While an appropriate increase in innovation inputs can enhance innovation outputs, surpassing a certain threshold may yield no additional benefits. Instead, it can escalate innovation-related expenditures and operational costs for the enterprise.

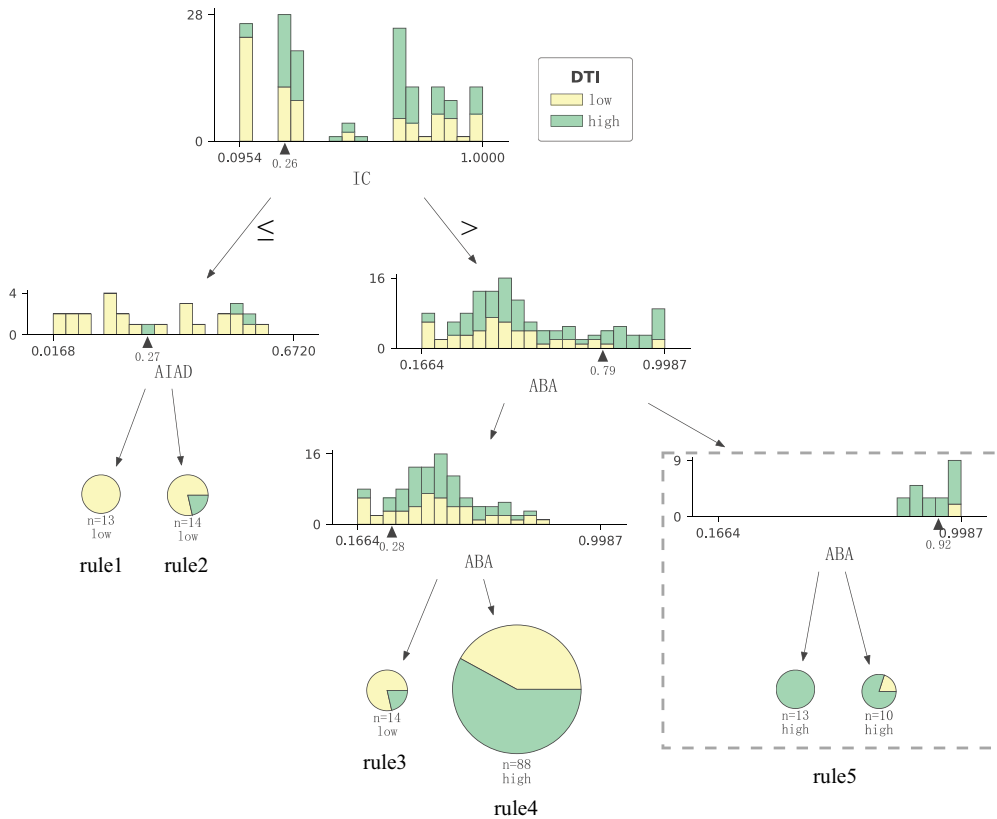
#### **4.4.2. Decision tree of flexible enterprises**

In flexible enterprises, industry concentration, AI adoption degree, and absorbing ability impact digital technology innovation. Moreover, varying combinations of these factors exert distinct effects on the performance of digital technology innovation. The decision tree pertaining to flexible enterprises is illustrated in [Figure 8](#).



**Figure 7.** Intelligent enterprises decision tree.

As illustrated in Figure 8, the left subtree of the flexible enterprise decision tree consists of two nodes: industry concentration and the AI adoption degree. When enterprises exhibit low industry concentration, they are likely to experience a low level of digital technology innovation. However, if the degree of AI adoption is increased, the probability of these enterprises achieving high levels of digital technology innovation also rises. Conversely, the right subtree comprises two nodes: industry concentration and absorbing ability. In environments characterised by high industry concentration, if a flexible enterprise exhibits weak absorbing ability, it is predisposed to attain low levels of digital technology innovation. Conversely, if its absorbing ability is medium, there is a tendency to achieve high levels of digital technology innovation. Furthermore, excellent absorbing ability significantly enhances the likelihood of obtaining high levels of digital technology innovation. A horizontal comparison indicates that, within flexible enterprises, the level of high digital technology innovation is influenced by industry concentration and absorbing ability rather than the degree of AI adoption. In contrast, enterprises with low digital technology innovation are affected by industry concentration, the degree of AI adoption, and absorbing ability.



**Figure 8.** Flexible enterprises decision tree.

In general, low industry concentration correlates with intense competition among firms, leading to conservative strategic approaches and limited adoption of AI, resulting in a relatively lower level of digital technology innovation. Conversely, the implementation of AI can enhance firms' resilience against risks and foster improvements in digital technology innovation. In contrast, high industry concentration typically results in reduced competition, with many firms positioned as industry leaders exhibiting advanced digital technology innovation. At this stage, firms can be categorised into three tiers based on their absorbing ability. Those with low absorbing ability tend to achieve limited levels of digital technological innovation. Firms with medium absorbing ability generally attain a higher level of digital technology innovation. Finally, firms with strong absorbing ability are likely to experience significantly elevated levels of digital technology innovation.

#### 4.5. Decision rule analysis

Drawing from the decision tree analysis, a set of decision rules for digital technology innovation is meticulously crafted. Specifically, the AI adoption degree, dynamic capability, and industry concentration serve as antecedent condition variables, while the digital technology innovation of enterprises is designated as the decision rule variable. This

framework facilitates the examination of the mechanisms that enhance digital technology innovation in enterprises under varying conditions. The *Support* and *Confidence* of each decision rule are computed according to formula (4) and formula (5), respectively. Furthermore, the distribution of different categories of digital technology innovation within each decision rule is analysed. Detailed information regarding the decision rules is presented in Table 3.

As illustrated in Table 3, intelligent enterprises exhibit 4 distinct decision rules for digital technology innovation, whereas flexible enterprises demonstrate 5. From the comprehensive analysis, several insights can be drawn: (1) There are significant discrepancies in the factors influencing digital technology innovation across different enterprise types, underscoring the importance of classification discussions. (2) The AI adoption degree and the absorbing ability are pivotal factors affecting the digital technology innovation within enterprises. (3) An examination of the *Confidence* levels associated with the rules reveals that the *Confidence* of most decision rules exceeds 60%, indicating a robust fit and suggesting high interpretability of the research findings.

Based on a comprehensive analysis of the decision-making frameworks of intelligent enterprises, as illustrated in Table 3 and Figure 7, it is observed that the characteristics of technology and organisation mutually influence the digital technology innovation within these enterprises. In contrast, industry concentration does not appear to impact digital technology innovation. Furthermore, both innovation ability and adaptive ability show no significant effect on the digital technology innovation of intelligent enterprises. In these entities, when the absorbing ability does not exceed 0.471, a majority of enterprises exhibit low digital technology innovation. Conversely, when the absorptive capacity exceeds 0.471, the majority demonstrate high levels of digital technology innovation. This indicates that absorbing ability has a considerable positive effect on digital technology innovation within intelligent enterprises. The relationship between the AI adoption degree and digital technology innovation is complex and exhibits nonlinear characteristics, which needs to be discussed in different situations. When the absorbing ability is less than or equal to 0.471, there is a 75% probability of achieving high levels of digital

**Table 3.** Digital technology innovation decision rule table.

| Firm type               | IC     | ABA           | IA | ADA | AIAD   | Support (%) | Confidence (%) | DTI  | Proportion       |
|-------------------------|--------|---------------|----|-----|--------|-------------|----------------|------|------------------|
| Intelligent enterprises | –      | ≤0.471        | –  | –   | ≤0.505 | 8.82        | 75.00          | high | l: 25%<br>h: 75% |
|                         | –      | ≤0.471        | –  | –   | >0.505 | 63.97       | 65.52          | low  | l: 66%<br>h: 34% |
|                         | –      | >0.471        | –  | –   | ≤0.988 | 19.12       | 80.77          | high | l: 19%<br>h: 81% |
|                         | –      | >0.471        | –  | –   | >0.988 | 8.09        | 54.55          | low  | l: 55%<br>h: 45% |
| Flexible enterprises    | ≤0.264 | –             | –  | –   | ≤0.274 | 8.56        | 100.00         | low  | l: 100%<br>h: 0% |
|                         | ≤0.264 | –             | –  | –   | >0.274 | 9.21        | 78.57          | low  | l: 79%<br>h: 21% |
|                         | >0.264 | ≤0.284        | –  | –   | –      | 9.21        | 78.57          | low  | l: 79%<br>h: 21% |
|                         | >0.264 | (0.284,0.786] | –  | –   | –      | 57.89       | 57.95          | high | l: 42%<br>h: 58% |
|                         | >0.264 | >0.786        | –  | –   | –      | 15.13       | 91.30          | high | l: 9%<br>h: 91%  |

technology innovation if the AI adoption degree does not exceed 0.505. Conversely, if the AI adoption degree exceeds 0.505, the likelihood of low levels of digital technology innovation increases to 66%. In contrast, when the absorbing ability exceeds 0.471, an 81% likelihood of obtaining high digital technology innovation exists if the AI adoption degree is less than or equal to 0.988. In this scenario, if the AI adoption degree surpasses 0.988, the probability of low digital technology innovation rises to 55%. These findings suggest that enhancing the absorbing ability of enterprises, in conjunction with a moderate increase in the AI adoption degree, serves as a critical pathway for improving the digital technology innovation capabilities of intelligent enterprises.

The dynamic capability of intelligent enterprises are currently inadequate, despite a high AI adoption degree. Therefore, the AI adoption degree should be dynamically adjusted based on the specific conditions of the enterprises to maximise their inherent advantages. When an enterprise's absorbing ability is limited, the AI adoption degree should be moderately reduced to conserve costs and optimise the utilisation of limited technological resources, thereby fostering effective digital technological innovation. Conversely, when an enterprise exhibits strong absorbing ability, an increase in AI adoption degree is warranted to fully leverage advanced external technologies. This influx of new ideas and technologies enhances the enterprise's digital innovation efforts and bolsters its internal innovation capabilities. Furthermore, it is crucial to avoid an excessive pursuit of perfection and indiscriminate adoption of AI technologies, as this could detract from the enhancement of other resources and capabilities within the enterprise, ultimately hindering digital technological innovation.

Through a comprehensive analysis of decision rules in flexible enterprises, as illustrated in Table 3 and Figure 8, it is evident that the interrelated factors of technology, organisation, and environment significantly influence the advancement of digital technology innovation within enterprises. The decision tree identifies industry concentration as the root node, segmenting the sample space into two distinct subgroups: high and low industry concentration. According to the decision rule table, when industry concentration is less than or equal to 0.264 and the AI adoption degree does not exceed 0.274, the *Confidence* level for flexible enterprises exhibiting low digital technology innovation is 100%. This finding underscores the joint positive impact of industry concentration and AI adoption degree on digital technology innovation, with a supporting sample proportion of 8.56%. A low industry concentration indicates that enterprises operate in a highly competitive environment, such as in the pharmaceutical manufacturing or chemical raw materials and products sectors. Enterprises within these contexts that hesitate to embrace advanced AI technologies due to fear of failure may confront the risks of innovation exhaustion, market marginalisation, or even elimination.

As illustrated in Figure 8, rule 1 shows that the proportion of high digital technology innovation is 0%. In contrast, rule 2 indicates an increase in this proportion to 21%. This suggests that enterprises operating in low-concentration industries may enhance their likelihood of achieving high digital technology innovation through the appropriate implementation of AI technology.

When industry concentration exceeds 0.264, three distinct categories emerge based on varying levels of absorbing ability, referred to as rule 3 through rule 5. In rule 3, enterprises with an absorbing ability of less than or equal to 0.284 exhibit a low *Confidence* level of 78.57% in their digital technology innovation. Rule 4 pertains to companies with an

absorbing ability ranging from 0.284 to 0.786, categorised as having medium absorbing ability. These enterprises demonstrate an increased propensity for high-level digital technology innovation, with a *Confidence* level of 57.95%. Finally, rule 5 encompasses enterprises with an absorbing ability exceeding 0.786, indicating a relatively high capacity. In this scenario, the likelihood of achieving high-level digital technology innovation is significantly enhanced, with *Confidence* rising to 91.30%.

The AI adoption among flexible enterprises remains limited. However, these organisations possess substantial dynamic capabilities. To address their weaknesses, they should leverage the strengths of others based on their specific circumstances. In low-industry concentration contexts, flexible enterprises are advised to strategically implement AI technologies to capitalise on the advantages offered by these innovations, thereby gaining a competitive edge in a challenging environment. Conversely, in high-concentration industries – such as food manufacturing, specialised equipment manufacturing, and automotive manufacturing – where many firms hold significant market leadership, flexible enterprises must maximise their absorbing ability. This involves implementing incentive measures to enhance employee motivation, fostering internal innovation potential, and actively collaborating with other institutions to facilitate the exchange of knowledge and skills. By effectively utilising both internal and external human and material resources, they can optimise overall benefits.

The AI adoption degree exhibits a complex nonlinear relationship with digital technology innovation, which necessitates an exploration of its interplay with dynamic capability and industry concentration. The mechanisms influencing enterprise digital technology innovation vary across different contexts. A case study by Trocin et al. (2021) revealed that higher levels of AI adoption degree positively affect the processes, services, and outcomes of digital innovation within enterprises. This finding aligns with previous research suggesting that intelligent enterprises should strategically enhance their AI adoption degree to foster digital technology innovation. This paper investigates the synergistic effects of organisational and environmental factors in conjunction with AI adoption, employing the TOE framework for a more comprehensive analysis. Research indicates that the integration of AI can significantly enhance innovation within China's manufacturing sector, albeit with varying marginal effects across different industries (Wang, Xiao, and Deng 2023). This echoes the two variables of artificial intelligence adoption degree and industry concentration in this paper, but the difference is that this paper focuses on the enterprise level, takes dynamic capabilities into account, and analyzes the nonlinear impact of artificial intelligence adoption degree on digital technology innovation in a fine-grained manner. Furthermore, individual-level studies reveal that employees' apprehensions regarding potential job displacement due to AI can adversely affect corporate digital innovation performance (Arias-Pérez and Vélez-Jaramillo 2022). This finding parallels the conclusions drawn in this paper, which suggest that excessive reliance on AI technology may undermine enterprise digital technology innovation. However, this study approaches the issue from an enterprise-level perspective rather than the psychological perspective of employees. A study examining the Malaysian manufacturing sector indicates that smart technology serves as a mediator in the relationship between digital transformation and operational performance. Consequently, manufacturing organisations are advised to prioritise the integration of smart technology within their supply chain processes to enhance operational performance (Lee et al. 2024). This finding is related to the



conclusions of this paper, which underscores the critical role of intelligent technology adoption in the digitalisation of manufacturing enterprises. However, this paper uniquely emphasises the adoption of AI, distinguishing it from previous research.

It is noteworthy that specific characteristics, such as innovation ability and adaptive ability, do not demonstrate a significant positive or negative effect in the scope of this study. This outcome can be ascribed to two principal reasons. Firstly, the decision tree's pruning process methodically eliminates rules that have minimal sample representation or low *Confidence* scores. Secondly, within the organisational dynamic capability framework, innovation ability and adaptive capacity are often eclipsed by the more dominant factors of absorptive capacity and technical and environmental considerations. Moreover, the even spread of these two indicators across different enterprises dilutes their overall impact on the decision-making rules.

## 5. Discussion

### 5.1. Conclusions

This study conducts an in-depth analysis of A-share manufacturing companies listed on the Shanghai and Shenzhen stock exchanges from 2021 to 2022. It employs a suite of machine learning techniques including cloud model calibration, K-Means clustering, and CART decision tree analysis to investigate the synergistic effects of the AI adoption degree, dynamic capability, and industry concentration on enterprise digital technology innovation within the TOE framework. The findings are as follows:

- (1) The cluster analysis indicates that the sample enterprises can be classified into two distinct types: intelligent enterprises and flexible enterprises, each exhibiting different characteristics. Intelligent enterprises demonstrate a high average AI adoption degree alongside low dynamic capability. Conversely, flexible enterprises showcase a high dynamic capability but possess low average AI adoption degree. Notably, there is no significant difference between the two types of enterprises regarding the average level of industry concentration.
- (2) Decision tree analysis indicates that, in intelligent enterprises, the AI adoption degree significantly influences digital technology innovation in a complex non-linear manner. Furthermore, the absorbing ability of enterprises positively affects their ability to innovate digitally. Specifically, when an enterprise's absorbing ability is low, reducing AI adoption can facilitate digital technology innovation. Conversely, when an enterprise demonstrates high absorbing ability, a greater degree of AI adoption correlates with enhanced digital technology innovation. However, it is important to exercise caution regarding excessive AI adoption, as it may lead to a lower level of digital technology innovation.
- (3) In flexible enterprises, the AI adoption degree, industry concentration, and absorbing ability positively influence digital technology innovation. At low levels of industry concentration, enterprises can enhance their AI adoption degree, thereby increasing the likelihood of achieving significant digital technology innovation. Conversely, in industries with high concentration, a stronger absorbing ability is associated with a greater potential for attaining substantial digital technology innovation.

- (4) The relationship between innovation ability, adaptive ability, and digital technology innovation is not significant. This lack of correlation among these characteristic variables regarding enterprise digital technology innovation may be obscured by other factors, leading to minimal informational discrepancies.

## **5.2. Implications**

Firstly, enterprises should gain a clear understanding of their current situation and strategically determine their AI adoption degree accordingly. It is essential to choose an appropriate trajectory for digital technology innovation. For instance, intelligent enterprises with lower absorptive capacity should carefully calibrate their AI adoption to ensure cost-effectiveness. On the other hand, enterprises with robust absorptive capacity should consider increasing their AI adoption degree to enhance the integration and management of innovative resources, both externally and internally, thus strengthening their internal drive for digital innovation. Moreover, companies should exercise caution against over-reliance on robots and AI technologies. Excessive dependence on automated decision-making processes can lead to a misleading belief that AI possesses the ability to generate knowledge independently. Instead, AI should be regarded as a supplementary tool, ensuring that it does not supplant human knowledge, reasoning, and experience. This approach will facilitate the advancement of digital technology innovation within enterprises.

In industries characterised by low concentration, such as pharmaceutical manufacturing and the production of chemical raw materials and products, flexible enterprises should strategically adopt AI technologies to establish a competitive edge in a highly competitive environment. Conversely, in industries with high concentration, such as food manufacturing, specialised equipment manufacturing, and automobile manufacturing, flexible enterprises must focus on enhancing their absorbing ability. This involves stimulating employees' innovative potential, fostering connections with upstream and downstream enterprises, universities, research institutions, suppliers, and customers, as well as actively assimilating both internal and external knowledge to elevate the level of digital technology innovation within the organisation.

Secondly, the government should consider initiating programmes and allocating funding to foster the development of AI technology. For instance, a robotics innovation platform could be established to consolidate knowledge resources from universities, enterprises, and R&D institutions, thereby enhancing the flow, spillover, and generation of knowledge. By integrating production, academia, and research, this initiative will bolster the independent innovation capabilities of enterprises in digital technology and steadily advance the innovation-driven development strategy that emphasises scientific and technological self-reliance. Simultaneously, the government must strengthen legislation and enforcement related to intellectual property rights, rigorously uphold intellectual property laws and regulations, and cultivate a conducive legal environment for enterprises to innovate in digital technology.

## **5.3. Limitations and future directions for research**

The data utilised in this study are cross-sectional, capturing a snapshot of the state of enterprise digital technology innovation and AI adoption. While this approach accounts

for the lag effects of innovation, it may not fully represent the dynamic evolution over time. Future research could integrate temporal dimensions to trace the progression and changes in the relationship between AI adoption and digital technology innovation more accurately. Expanding the dataset to encompass multiple years would also enhance the findings' applicability and generalisability.

This paper delves into the impact of the AI adoption on corporate digital technology innovation. Additionally, subsequent studies could investigate the determinants of digital technology innovation through alternative research frameworks and perspectives. It would also be beneficial to explore the effects of other digital technology adoptions on innovation, comparing them to AI adoption to determine which technology has a more significant impact on digital technology innovation.

## Disclosure statement

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